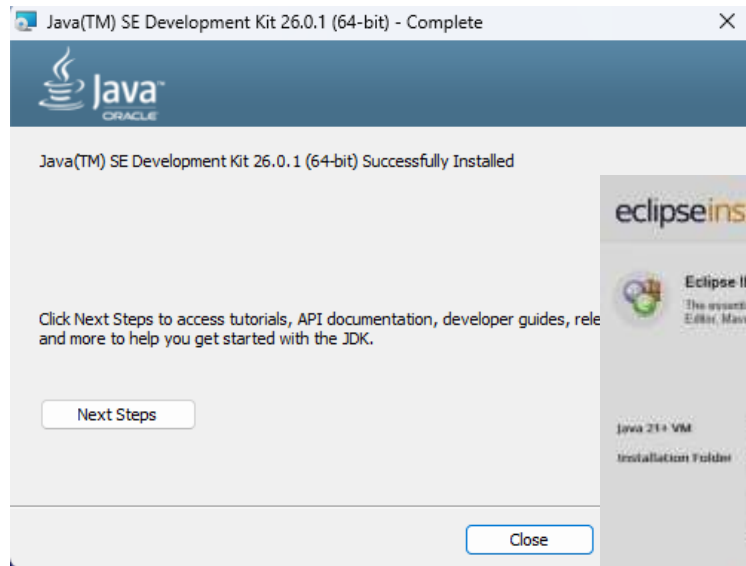


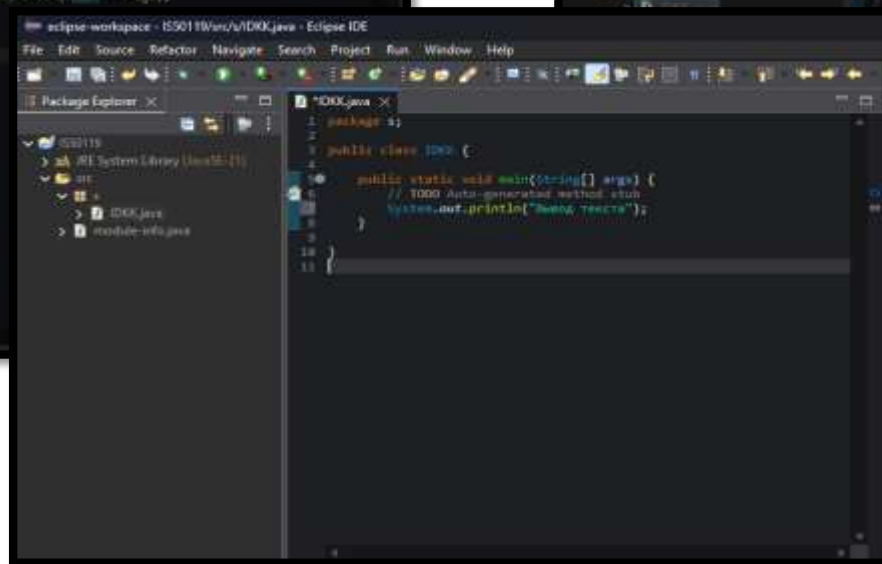
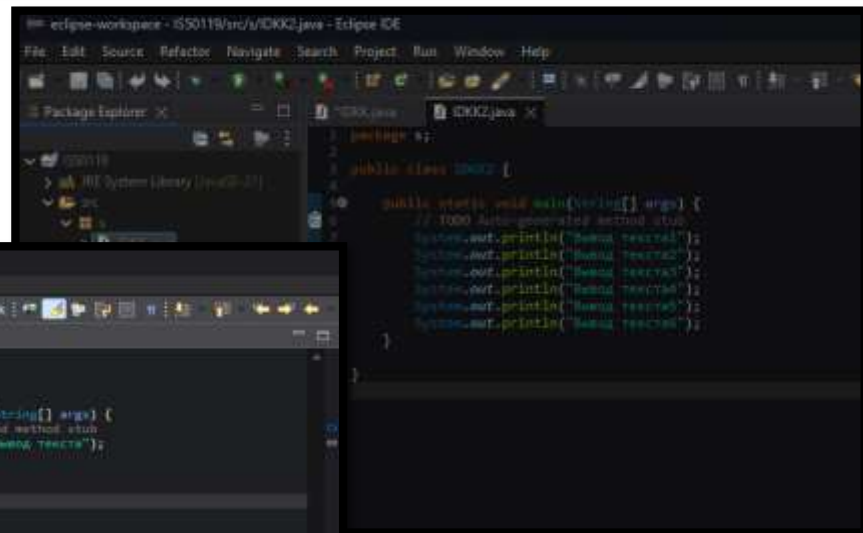
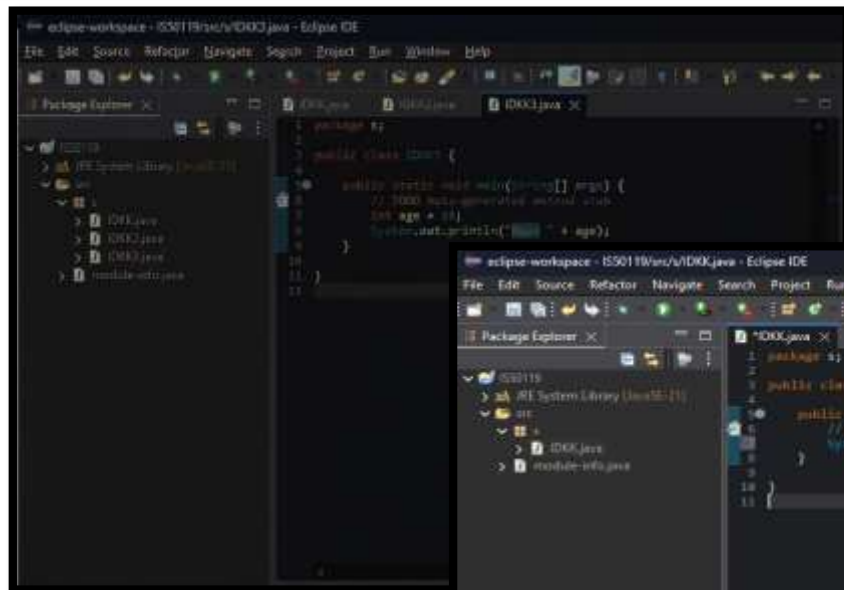
Отчет по лабораторным работам

Выполнил курсант 331 г
Полубудкин Ярослав

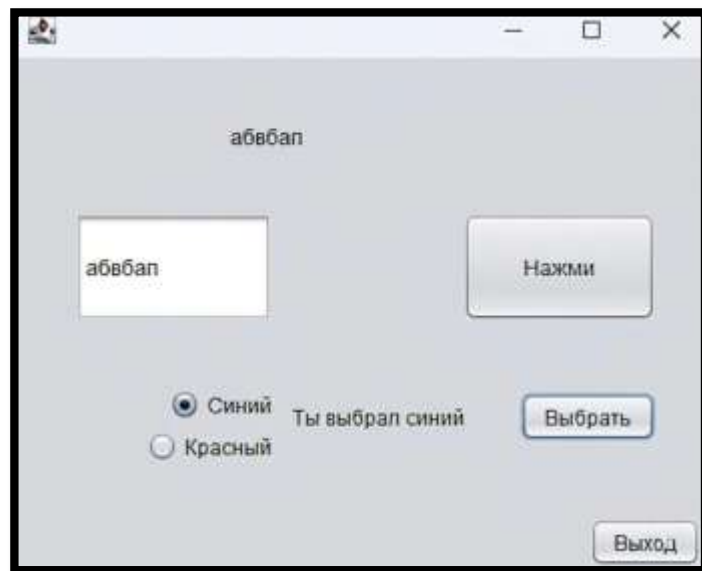
Подготовка рабочей среды и установка необходимого программного обеспечения.



Разработка и запуск стартового проекта в среде разработки Eclipse IDE.



Создание первичного приложения в NetBeans IDE.



Программная реализация приложения «Калькулятор».

```
/*
 * Click https://ibmhost/SystemFileSystem/Template/License/license-default.txt to change this
 * Click https://ibmhost/SystemFileSystem/Template/Classes/Main.java to edit this template
 */
package calculator;
import java.util.Scanner;

public class Calculator {

    public static void main(String[] args) {
        System.out.println("Введите операцию: ");
        System.out.println("1. Сложение");
        System.out.println("2. Вычитание");
        System.out.println("3. Умножение");
        System.out.println("4. Деление");
        //Введите значение
        Scanner scanner = new Scanner(System.in);
        int operation = scanner.nextInt();

        System.out.println("Введите первое число: ");
        int x = scanner.nextInt();
        System.out.println("Введите второе число: ");
        int y = scanner.nextInt();

        int result = 0;

        if (operation == 1)
            result = x + y;
        else if (operation == 2)
            result = x - y;
        else if (operation == 3)
            result = x * y;
        else if (operation == 4)
            result = x / y;

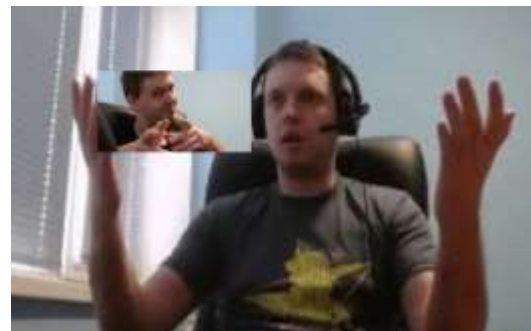
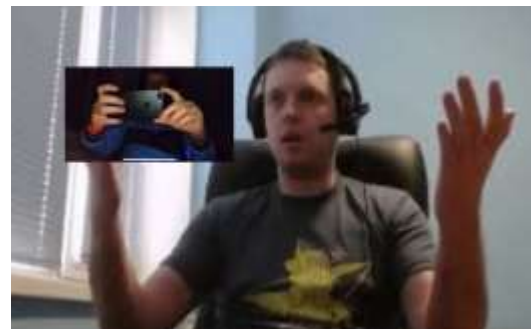
        System.out.println("Результат = " + result);
        System.out.println("Конец");
    }
}
```

Финальный этап: разработка двух самостоятельных программных модулей.

```
Game.java
Source History
1 import java.awt.Graphics;
2 import javax.swing.JFrame;
3 import javax.swing.JPanel;
4 import javax.swing.WindowConstants;
5
6 public class Game extends JFrame {
7
8     private static Game game;
9
10    public static void main(String[] args) {
11        game = new Game();
12        game.setDefaultCloseOperation(WindowConstants.EXIT_ON_CLOSE);
13        game.setLocation(200, 50);
14        game.setSize(900, 600);
15        game.setResizable(false);
16        GameField game_field = new GameField();
17        game.add(game_field);
18        game.setVisible(true);
19    }
20
21    public static void onRepaint(Graphics g) {
22        g.fillOval(10, 10, 200, 100);
23    }
24
25    public static class GameField extends JPanel {
26        @Override
27        protected void paintComponent(Graphics g) {
28            super.paintComponent(g);
29            onRepaint(g);
30        }
31    }
32
33 }
```



```
Game.java
Source History
8 import java.awt.event.MouseEvent;
9 import java.io.IOException;
10 import javax.swing.ImageIcon;
11 import javax.swing.JFrame;
12 import javax.swing.WindowConstants;
13
14 public class Game extends JFrame {
15
16     private static Data data_game;
17     private static long start_time_game;
18     private static Image image_bg;
19     private static Image image_enemy;
20     private static Image image_enemy2;
21     private static float time_game = 0.0f;
22     private static float time_enemy = 0.0f;
23     private static float time_enemy2 = 0.0f;
24     private static int score = 0;
25
26     public static void main(String[] args) throws IOException {
27         ImageIO.read(new File(getClass().getResourceAsStream("game_bg.png")));
28         ImageIO.read(new File(getClass().getResourceAsStream("game_enemy.png")));
29         ImageIO.read(new File(getClass().getResourceAsStream("game_enemy2.png")));
30         data_game = new Game();
31         data_game.setDefaultCloseOperation(WindowConstants.EXIT_ON_CLOSE);
32         data_game.setSize(800, 600);
33         data_game.setTitle("Game");
34         data_game.setResizable(false);
35         data_game.setVisible(true);
36         data_game.setLocationRelativeTo(null);
37         GameField game_field = new GameField();
38         game_field.addMouseListener(new MouseAdapter() {
39             @Override
40             public void mouseClicked(MouseEvent e) {
41                 int x = e.getX();
42                 int y = e.getY();
43                 float drop_right = data_game.getDropRight();
44                 float drop_left = data_game.getDropLeft();
45                 boolean is_drop = x >= drop_right && y >= drop_bottom && x <= drop_left && y <= drop_top;
46
47                 if (is_drop) {
48                     drop_top = -100;
49                     drop_left = (float) Math.random() * (game_field.getWidth() - data_game.getWidth());
50                     drop_right = (float) Math.random() * (game_field.getWidth() - data_game.getWidth());
51                     drop_bottom = 100;
52                 }
53             }
54         });
55     }
56 }
```



Инициализация локального репозитория Git в качестве завершающего шага контроля версий.

```
Командная строка
Microsoft Windows [Version 10.0.26200.8246]
(c) Корпорация Майкрософт (Microsoft Corporation). Все права защищены.

C:\Users\TheIW>cd C:\Users\TheIW\Desktop\Лаб 1-6

C:\Users\TheIW\Desktop\Лаб 1-6>git init
Initialized empty Git repository in C:/Users/TheIW/Desktop/Лаб 1-6/.git/

C:\Users\TheIW\Desktop\Лаб 1-6>|
```